Electrowinning of Copper

UMT302L Materials Processing Laboratory

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Bath assigned: 10 g $CuSO_4 \cdot 5H_2O$/L, 0.1 M H_2SO_4
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1 Objective

- Quantify the minimum Cu concentration at which Cu can be recovered by electrowinning.
- Determine the effect of cathode size (2×5 cm vs 4×5 cm) on recovery time and mass of Cu recovered.

2 Theory

2.1 Industrial Context

Electrowinning is the final step of the hydrometallurgical route for copper, commonly called SX–EW (solvent extraction–electrowinning). Low-grade oxide ores and some secondary sulphides are heap- or dump-leached with dilute sulphuric acid. The pregnant leach solution is too dilute and impure for direct electrolysis, so Cu^{2+} is selectively transferred into an organic extractant by solvent extraction and then stripped into a clean, concentrated sulphate electrolyte. This strip liquor is fed to the electrowinning cells. SX–EW accounts for roughly a fifth of global primary copper output and is favoured where ore grade, scale or the cost of smelting and transport rule out the pyrometallurgical route.

In an industrial tankhouse, the cathode and anode reactions are

$$\text{Cathode:} \quad Cu^{2+} + 2e^- \rightarrow Cu \qquad E^\circ = +0.34 \text{ V}, \tag{1}$$

$$\text{Anode:} \quad 2H_2O \rightarrow O_2 + 4H^+ + 4e^- \qquad E^\circ = +1.23 \text{ V}, \tag{2}$$

giving the overall reaction

$$2CuSO_4 + 2H_2O \rightarrow 2Cu + 2H_2SO_4 + O_2. \tag{3}$$

Unlike electrorefining, the anode is inert and does not supply copper. Copper is depleted from the electrolyte and acid is regenerated, so the spent electrolyte is returned to the leaching circuit.

Typical operating conditions are summarised in Table 1. These are representative ranges and vary between plants.

Table 1: Typical industrial copper electrowinning conditions.

Parameter	**Typical value**
Cu^{2+} in electrolyte	30–50 g/L
H_2SO_4	150–180 g/L
Temperature	40–50 °C
Current density	200–400 A/m^2
Cell voltage	1.8–2.2 V
Current efficiency	90–95%
Specific energy consumption	∼2 kWh/kg Cu
Cathode purity	≥99.99% Cu

Cathodes are usually stainless steel (316L) permanent blanks, from which the deposit is stripped periodically, or copper starter sheets. Anodes are lead alloys (Pb–Ca–Sn), which are stable in the acidic, oxygen-evolving environment. Small additions of cobalt sulphate stabilise the anode film and reduce lead contamination and corrosion. Organic additives such as guar gum promote a smooth, dense deposit. Chloride is held at a low level to prevent pitting of the stainless steel cathodes and anode attack. Because the cell voltage is far above the reversible value, a large share of the energy goes into the oxygen evolution overpotential and ohmic losses, and current efficiency is lowered by Fe^{3+}/Fe^{2+} cycling and stray currents.

The bath used in this experiment (10 g $CuSO_4$/L in 0.1 M H_2SO_4) is considerably more dilute than industrial electrolyte, so mass-transport limitation and hydrogen evolution at the cathode are more significant than in practice.

Electrode reactions.

$$\text{Cathode:} \quad Cu^{2+} + 2e^- \rightarrow Cu \tag{4}$$

$$\text{Anode:} \quad 2H_2O \rightarrow O_2 + 4H^+ + 4e^- \tag{5}$$

Faraday's laws and kinetics.

$$m_{\text{th}} = \frac{M\,I\,t}{n\,F} \tag{6}$$

Electrowinning vs cementation. Both processes recover Cu^{2+} from solution as Cu metal, but the source of electrons differs. Cementation is a spontaneous galvanic displacement, $Cu^{2+} + Fe \rightarrow Cu + Fe^{2+}$, driven by the potential difference between $E^\circ_{Cu^{2+}/Cu} = +0.34$ V and $E^\circ_{Fe^{2+}/Fe} = -0.44$ V ($\Delta E^\circ \approx 0.78$ V). It needs no external power, but it consumes a sacrificial reductant (nearly 1 mol Fe per mol Cu), contaminates the solution with Fe^{2+}, and gives a loosely adherent, impure cement that needs further refining. Its rate is controlled by mass transport to the Fe surface and by the available surface area.

Electrowinning uses an external power supply to drive a non-spontaneous overall reaction, $Cu^{2+} + H_2O \rightarrow Cu + \frac{1}{2}O_2 + 2H^+$, with $E^\circ_{\text{cell}} \approx -0.89$ V. It consumes only electricity, regenerates acid at the anode, and gives a dense, high-purity cathode. The rate is set directly by the applied current through Faraday's law, so it is easy to control. The trade-off is energy and capital cost. Cementation therefore suits dilute, low-value streams, while electrowinning suits concentrated leach liquors where product purity matters.

Effect of other metals / other target metals. A real leach solution contains several metal ions, and which of them deposit depends on their reduction potentials and overpotentials. Ions nobler than Cu (Ag^+, Au^{3+}) deposit first and contaminate the product. Ions with potentials close to Cu or slightly lower (Pb, Sn, Ni at -0.13, -0.14, -0.25 V) can co-deposit if the cathode potential is pushed too negative or the current density is too high. Fe^{3+} is a particular problem in Cu electrowinning: it is reduced at the cathode ($Fe^{3+} + e^- \rightarrow Fe^{2+}$, $E^\circ = +0.77$ V) and re-oxidized at the anode, a parasitic cycle that wastes current and lowers current efficiency. Impurities are controlled by solvent extraction or purification before electrowinning, by a bleed stream, and by holding the cathode potential within a window selective for the target metal.

For a different target metal, the competing reaction becomes hydrogen evolution. Zn ($E^\circ = -0.76$ V) can still be electrowon from aqueous sulfate because H_2 evolution on Zn is kinetically slow (high overpotential), but this requires very pure electrolyte, since traces of Cu, Co, Ni or Sb catalyse H_2 evolution and cause the deposit to redissolve. Al ($E^\circ = -1.66$ V) cannot be deposited from aqueous solution at all because H_2 evolves first; it is produced by electrolysis in molten cryolite (Hall–Héroult process) at a much higher energy cost.

Optimization and scaling. Optimization balances deposition rate, deposit quality and energy consumption per unit mass. Faraday's law gives a rate proportional to I, but this is bounded by the mass-transport-limited current,

$$i_L = \frac{nFDC_b}{\delta}, \tag{7}$$

where C_b is the bulk Cu^{2+} concentration, D the diffusion coefficient and δ the diffusion layer thickness. Above i_L the deposit becomes powdery or dendritic and H_2 evolution takes a growing share of the current, so current efficiency falls. This also sets the practical lower concentration limit examined in this experiment: at low C_b the usable current density drops. Energy per unit mass scales roughly as V_{cell}/η, and V_{cell} rises with current density (ohmic drop plus overpotentials), so there is an optimum rather than a simple maximum. Levers for improvement include higher temperature (larger D and conductivity), agitation or electrolyte circulation (smaller δ), a smaller electrode gap, levelling additives for smooth deposits, and a more conductive electrolyte.

Scale-up is achieved by numbering up rather than enlarging a single cell. Industrial tank houses use many parallel cathode and anode plates with flowing electrolyte, so the total current scales with plate area while the current density stays in the optimal range. Scale-up also requires information not obtainable from this laboratory: kinetics under flow, impurity tolerance, anode lifetime, and the economics of power, acid make-up and cathode handling.

This bath vs a real electrolyte. The bath used here is a simplified, unstirred, room-temperature, 200 mL Cu/H_2SO_4 solution containing 5–20 g/L $CuSO_4{\cdot}5H_2O$ (roughly 1.3–5 g/L Cu). A typical industrial copper electrowinning electrolyte contains about 30–50 g/L Cu and 150–180 g/L H_2SO_4, operates at roughly 45–50 °C at current densities of order 200–400 A/m^2, and is circulated continuously. It also contains additives (for example guar gum or similar levelling agents for smooth deposits, and cobalt to stabilize the anode), carefully controlled chloride, and dissolved Fe, Mn and other species that must be managed. Industrial cells use stainless steel cathodes from which the deposit is stripped and lead-alloy anodes, whereas here a brass cathode and an inert Ir anode are used.

These differences matter when interpreting the results. At the low concentrations and without stirring used here, the applied current density may exceed the limiting current, so a lower current efficiency and a rougher deposit than in industrial practice are expected. The brass cathode may also introduce error, since Zn in the brass can dissolve or take part in displacement reactions, which affects the mass balance.

3 Procedure

1. Prepared 200 mL of bath containing 10 g/L $CuSO_4{\cdot}5H_2O$ in 0.1 M H_2SO_4.
2. Cut two brass cathodes (2×5 cm and 4×5 cm); washed with soapy water, rinsed with DI water, dried in oven for $\sim$1 min.
3. Recorded initial masses
4. Marked each cathode at 2.5 cm to fix the immersed area (reproducible active area).
5. Connected brass to the working (cathode) terminal and the Ir electrode to the anode.
6. Applied constant current for 10 min: 100 mA (2×5 cm), 200 mA (4×5 cm).
7. Dried plated cathode in oven for 1–2 min; measured final mass.
8. Saved leftover electrolyte in a vial for AAS.

Observations. When electrodes are too close, the deposited metal can bridge the two electrodes
Due to vibrational disturbances we see some fluctuations in the graph
The surface layers of the deposited metal were too loosely held to the electrode and fell off before

we could take the mass measurements.
The graphs have a lot of noise in the starting, that is because we were trying to set up the electrodes into a stable position.

4 Data

Table 2: Cathode masses

Cathode	m_{initial} (g)	m_{final} (g)	Δm (g)
2×5 cm	0.4048	0.4266	0.0218
4×5 cm	0.8460	0.9280	0.0820

Other recorded quantities:
immersed area(for 2x5 brass electrode) = 4.9 cm^2 for 2x5 brass electrode
immersed area(for 4x5 brass electrode) = 9.9 cm^2 (for 4x5 brass electrode)

5 Analysis

5.1 Current and voltage vs time

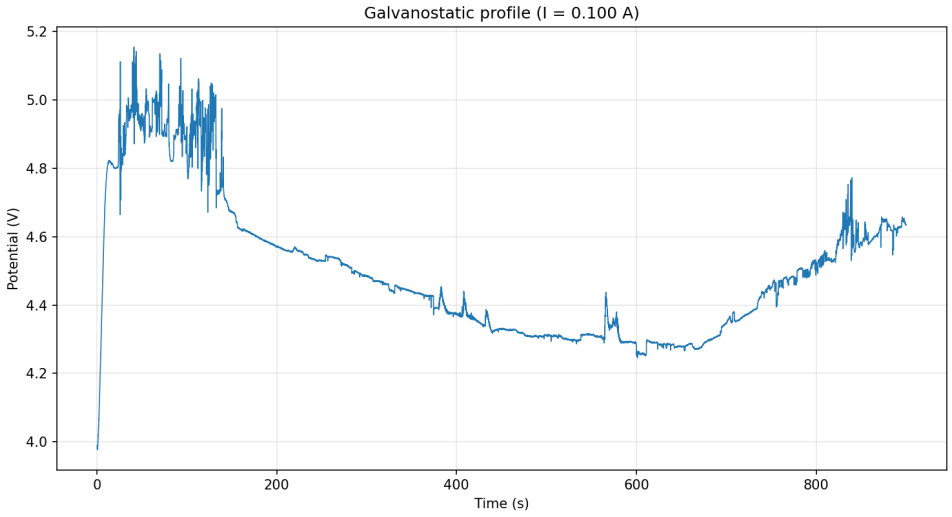


Figure 1: Current and voltage vs time for 2x5 electrode

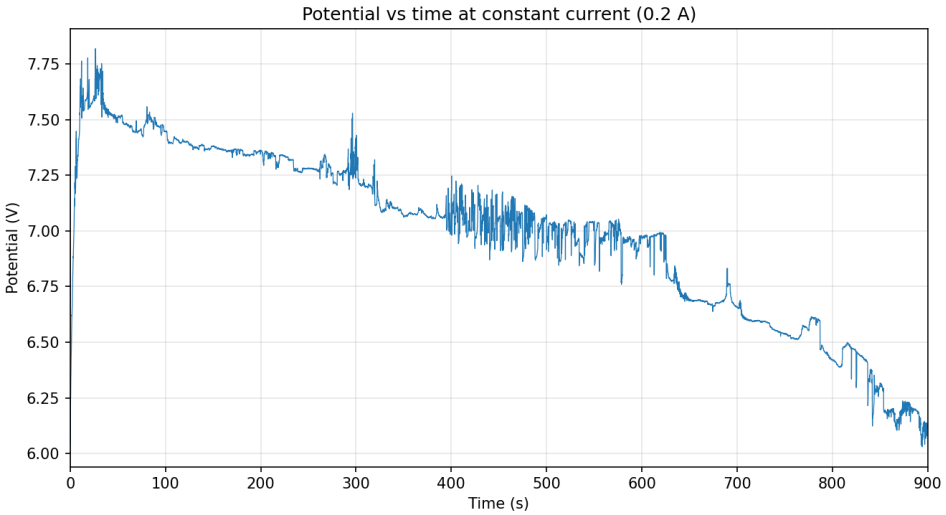


Figure 2: Current and voltage vs time for 4x5 electrode

5.2 Deposit area and current density

$$j = \frac{I}{A} \tag{8}$$

5.3 Mass of Cu recovered

$$\Delta m = m_{\text{final}} - m_{\text{initial}}, \qquad \delta(\Delta m) = \sqrt{\delta m_i^2 + \delta m_f^2} \tag{9}$$

5.4 Theoretical mass and current efficiency

With $M = 63.55$ g/mol, $n = 2$, $F = 96485$ C/mol:

$$m_{\text{th}} = \frac{M\, I\, t}{n\, F}, \qquad \eta = \frac{\Delta m}{m_{\text{th}}} \times 100\% \tag{10}$$

Discuss any shortfall (H_2 evolution, loosely adherent deposit loss, brass dissolution, weighing error).

5.5 Electrical energy and cost per gram

$$E = \int_0^t V(t')\, I(t')\, dt' \approx \sum_k V_k I_k\, \Delta t_k \tag{11}$$

Assumed tariff: [1.5cm]6.8 INR/kWh.

$$\text{Cost per g} = \frac{E_{\text{kWh}} \times \text{tariff}}{\Delta m} \tag{12}$$

5.6 Comparison with cementation

Cost per g (including consumed Fe), recovery, and time.

5.7 Comparison of cathodes

	2×5 cm	4×5 cm
I (mA)	100	200
j (mA/cm^2)	10	10
Δm (g)	0.0218	0.0820
η (%)	73.65	138.28
E (kWh)	0.000113	0.000349
Cost per g	0.029	0.035

6 Results

Optimal parameters by (1) cost per g and (2) mass recovered for a given time, with numbers.

Bath (g/L)	Cathode	Δm (g)	η (%)	Cost/g
10	2x5	0.0218	73.65	0.035
10	4x5	0.0820	138.28	0.029

7 Conclusions

We got more amount of copper than predicted by theory, that because the copper deposited on the cathode extended out from the cathode on the surface of the electrolyte and bridged to the other electrode this increased the current flow and therefore more copper got deposited than what was predicted by theory.
Due to this we got large errors and can't conclude if a smaller electrode is more economical or a larger electrode